Foundation of Programming

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Class: FY BTech CSE CsF Div-13

Assignment 5

Matrix Operations

```
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], c[10][10];
    int i, j, k, r1, c1, r2, c2, choice;

    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter elements of Matrix A:\n");
    for(i=0;i<r1;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&a[i][j]);

    printf("Enter rows and columns of Matrix B: ");
    scanf("%d %d", &r2, &c2);

    printf("Enter elements of Matrix B:\n");
    for(i=0;i<r2;i++)
        for(j=0;j<c2;j++)
            scanf("%d",&b[i][j]);

    printf("\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Transpose\n5.Saddle Point\n");
    printf("Enter choice: ");
    scanf("%d",&choice);

    switch(choice) {

        case 1: // Addition
```

```
    if(r1==r2 && c1==c2) {
        for(i=0;i<r1;i++) {
            for(j=0;j<c1;j++) {
                c[i][j] = a[i][j] + b[i][j];
                printf("%d ", c[i][j]);
            }
            printf("\n");
        }
    } else {
        printf("Addition not possible\n");
    }
    break;

case 2: // Subtraction
    if(r1==r2 && c1==c2) {
        for(i=0;i<r1;i++) {
            for(j=0;j<c1;j++) {
                c[i][j] = a[i][j] - b[i][j];
                printf("%d ", c[i][j]);
            }
            printf("\n");
        }
    } else {
        printf("Subtraction not possible\n");
    }
    break;

case 3: // Multiplication
    if(c1 == r2) {
        for(i=0;i<r1;i++) {
            for(j=0;j<c2;j++) {
                c[i][j] = 0;
                for(k=0;k<c1;k++)
                    c[i][j] += a[i][k] * b[k][j];
                printf("%d ", c[i][j]);
            }
            printf("\n");
        }
    } else {
        printf("Multiplication not possible\n");
    }
    break;

case 4: // Transpose of A
    for(i=0;i<c1;i++) {
        for(j=0;j<r1;j++) {
            printf("%d ", a[j][i]);
        }
```

```
            printf("\n");
        }
        break;

    case 5: // Saddle Point (for matrix A)
    {
        int min, pos, found = 0;

        for(i=0;i<r1;i++) {
            min = a[i][0];
            pos = 0;

            // find min in row
            for(j=1;j<c1;j++) {
                if(a[i][j] < min) {
                    min = a[i][j];
                    pos = j;
                }
            }

            // check if it's max in column
            for(k=0;k<r1;k++) {
                if(a[k][pos] > min)
                    break;
            }

            if(k == r1) {
                printf("Saddle point: %d\n", min);
                found = 1;
            }
        }

        if(!found)
            printf("No saddle point\n");

        break;
    }

    default:
        printf("Invalid choice");
  }

  return 0;
}
```

main.c

```c
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], c[10][10];
    int i, j, k, r1, c1, r2, c2, choice;

    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter elements of Matrix A:\n");
    for(i=0;i<r1;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&a[i][j]);

    printf("Enter rows and columns of Matrix B: ");
    scanf("%d %d", &r2, &c2);

    printf("Enter elements of Matrix B:\n");
    for(i=0;i<r2;i++)
        for(j=0;j<c2;j++)
            scanf("%d",&b[i][j]);

    printf("\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Transpose\n5.Saddle Point\n");
    printf("Enter choice: ");
```

Output

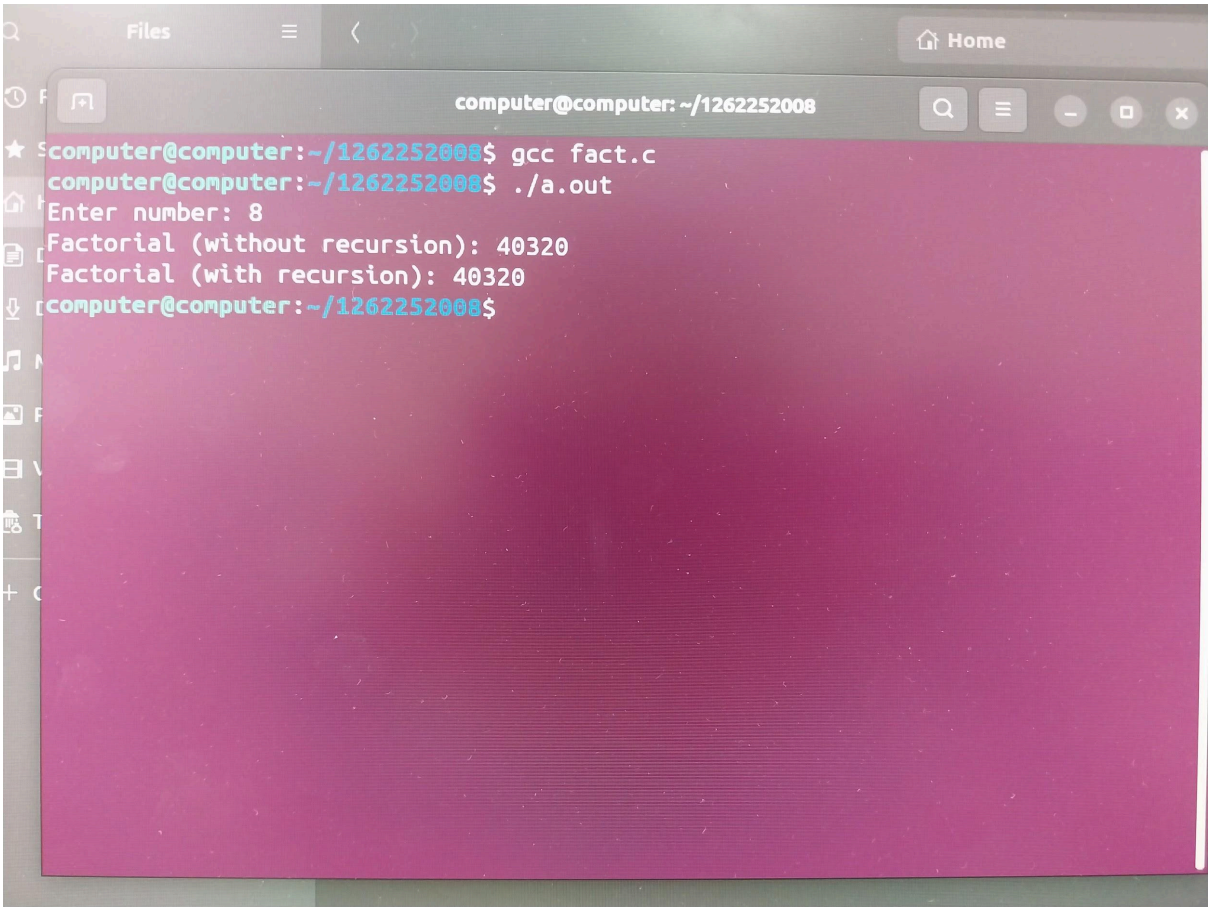
```
Enter rows and columns of Matrix A: 2 2
Enter elements of Matrix A:
1 2 3 4
Enter rows and columns of Matrix B: 2 2
Enter elements of Matrix B:
5 6 7 8

1.Addition
2.Subtraction
3.Multiplication
4.Transpose
5.Saddle Point
Enter choice: 4
1 3
2 4


=== Code Execution Successful ===
```

Assignment 6

Factorial (with & without recursion)

```c
#include <stdio.h>

int fact1(int n) {
    int i, f = 1;
    for(i = 1; i <= n; i++)
        f *= i;
    return f;
}

int fact2(int n) {
    if(n == 0 || n == 1)
        return 1;
    return n * fact2(n - 1);
}

int main() {
    int n;
    printf("Enter number: ");
    scanf("%d", &n);

    printf("Factorial (without recursion): %d\n", fact1(n));
    printf("Factorial (with recursion): %d\n", fact2(n));

    return 0;
```

```
}
```

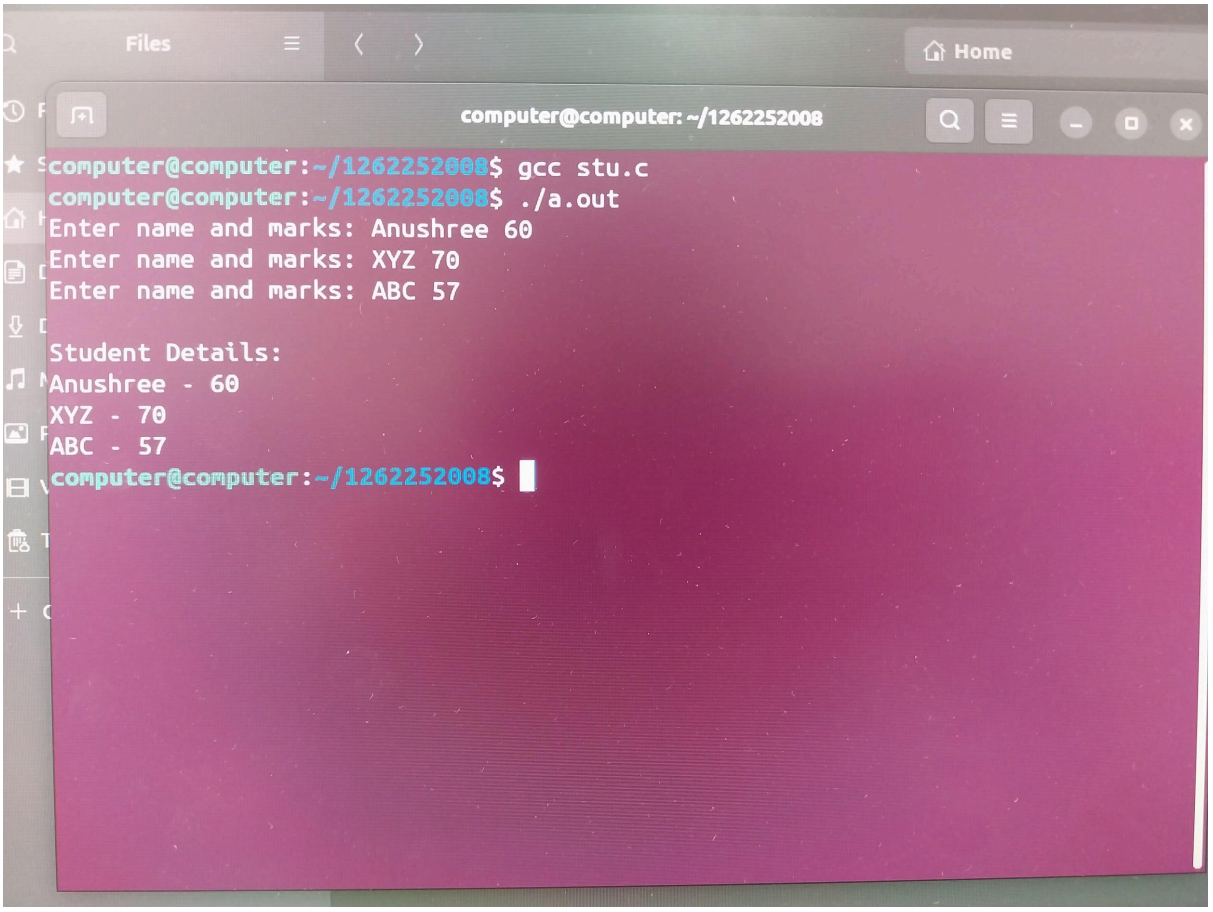


Assignment 7:

Student details using array of structures

```
#include <stdio.h>

struct student {
    char name[50];
    int marks;
};

int main() {
    struct student s[3];
    int i;

    for(i = 0; i < 3; i++) {
        printf("Enter name and marks: ");
        scanf("%s %d", s[i].name, &s[i].marks);
    }
```

```
    printf("\nStudent Details:\n");
    for(i = 0; i < 3; i++) {
        printf("%s - %d\n", s[i].name, s[i].marks);
    }

    return 0;
}
```

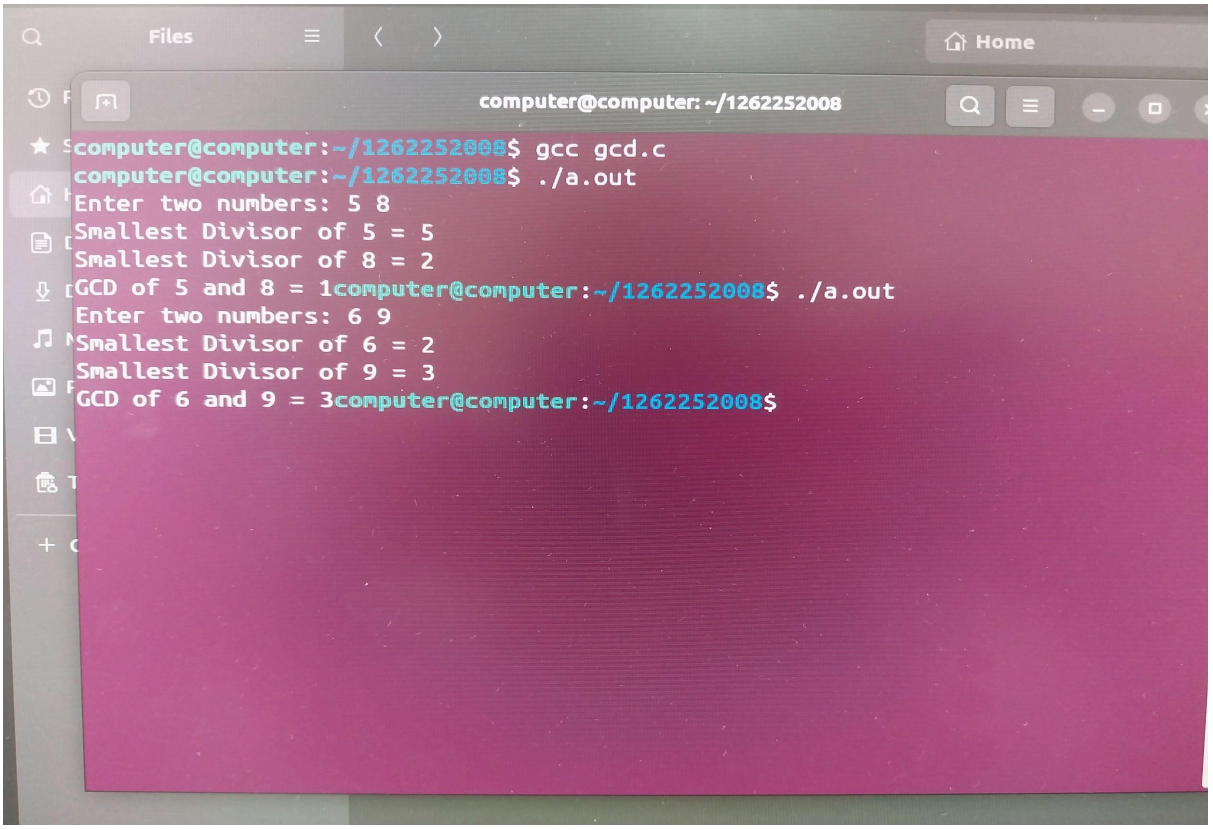


Assignment 12:

GCD & smallest divisor

```
#include <stdio.h>
// Function to find smallest divisor
int smallestDivisor(int n) {
int i;
if (n <= 1)
return n;
for (i = 2; i <= n; i++) {
if (n % i == 0)
```

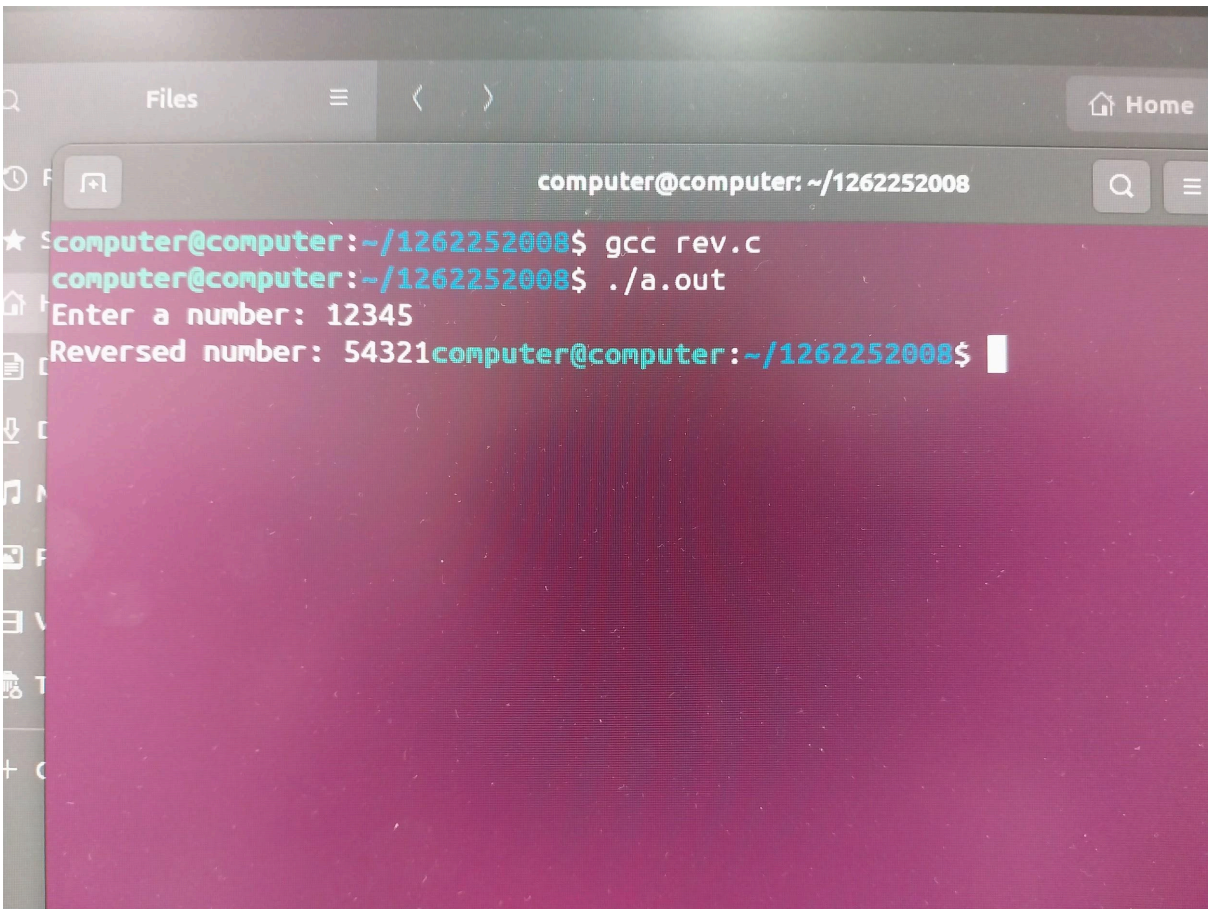
```
return i;
}
return n;
}
// Function to find GCD using Euclidean algorithm
int gcd(int a, int b) {
int temp;
while (b != 0) {
temp = b;
b = a % b;
a = temp;
}
return a;
}
int main() {
int num1, num2;

printf("Enter two numbers: ");
scanf("%d %d", &num1, &num2);
printf("Smallest Divisor of %d = %d\n", num1,
smallestDivisor(num1));
printf("Smallest Divisor of %d = %d\n", num2,
smallestDivisor(num2));
printf("GCD of %d and %d = %d", num1, num2, gcd(num1,
num2));
return 0;
}
```



Assignment 13:

Reverse digits

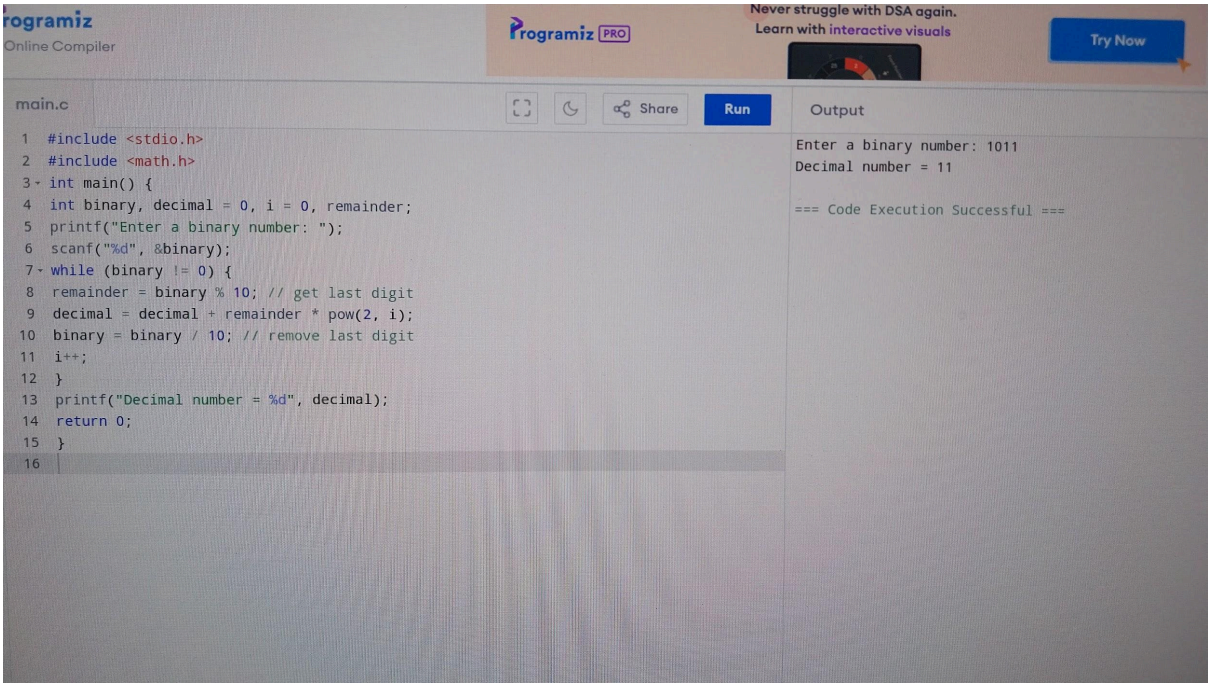
```
#include <stdio.h>
int main() {
 int num, digit;
 printf("Enter a number: ");
 scanf("%d", &num);
 printf("Reversed number: ");
 while(num != 0) {
 digit = num % 10;
 printf("%d", digit);
 num = num / 10;
 }
 return 0;
}
```



Assignment 14:

Binary to decimal

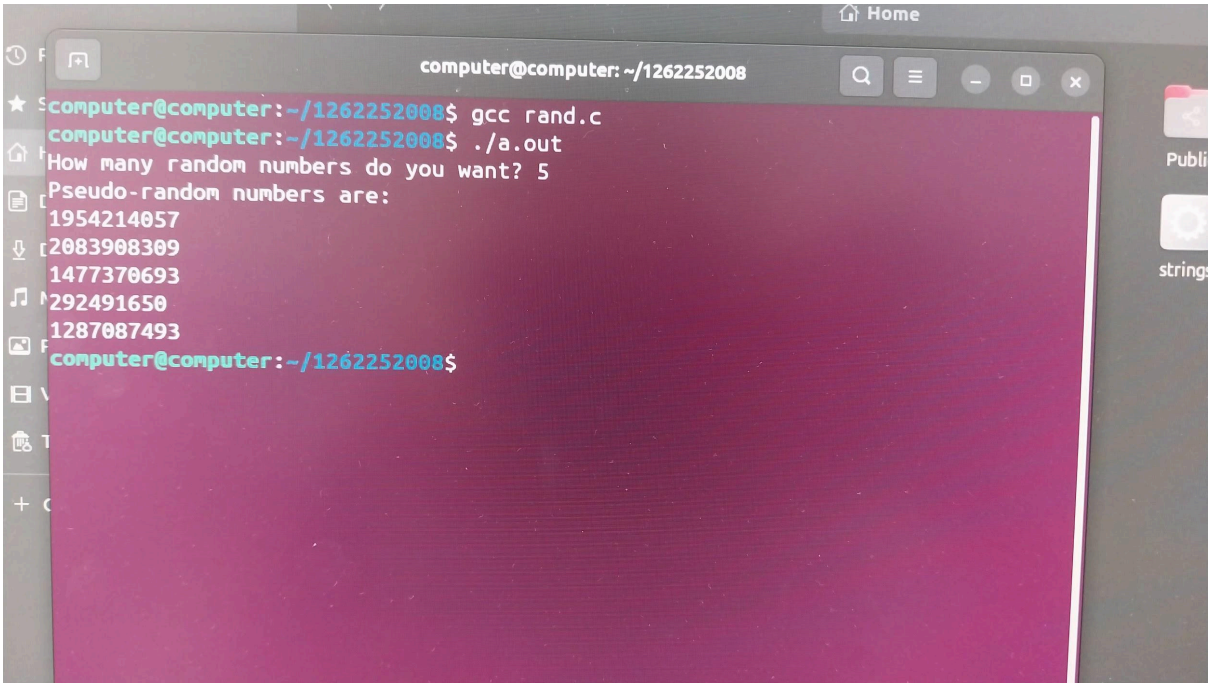
```
#include <stdio.h>
#include <math.h>
int main() {
int binary, decimal = 0, i = 0, remainder;
printf("Enter a binary number: ");
scanf("%d", &binary);
while (binary != 0) {
remainder = binary % 10; // get last digit
decimal = decimal + remainder * pow(2, i);
binary = binary / 10; // remove last digit
i++;
}
printf("Decimal number = %d", decimal);
return 0;
}
```



Assignment 15:

Random numbers

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
int main() {
int i, n;
printf("How many random numbers do you want? ");
scanf("%d", &n);
srand(time(0)); // seed using current time
printf("Pseudo-random numbers are:\n");
for(i = 0; i < n; i++) {
printf("%d\n", rand());
}
return 0;
}
```

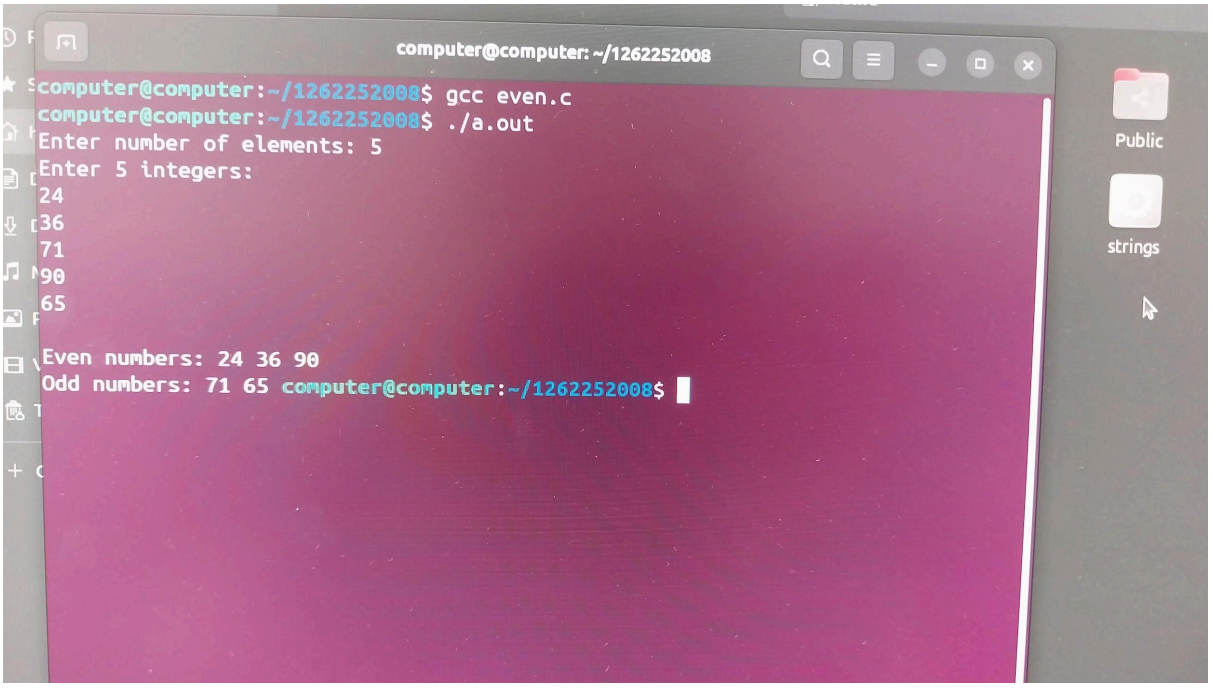


Assignment 16:

Even and odd

```
#include <stdio.h>
int main() {
int n, i;
int arr[100], even[100], odd[100];
int e = 0, o = 0;
printf("Enter number of elements: ");
scanf("%d", &n);
printf("Enter %d integers:\n", n);
for(i = 0; i < n; i++) {
scanf("%d", &arr[i]);
if(arr[i] % 2 == 0) {
even[e++] = arr[i]; // store in even array
} else {
odd[o++] = arr[i]; // store in odd array
}
}

// Display even numbers
printf("\nEven numbers: ");
for(i = 0; i < e; i++) {
printf("%d ", even[i]);
}
printf("\nOdd numbers: ");
for(i = 0; i < o; i++) {
```

```
printf("%d ", odd[i]);
}
return 0;
}
```

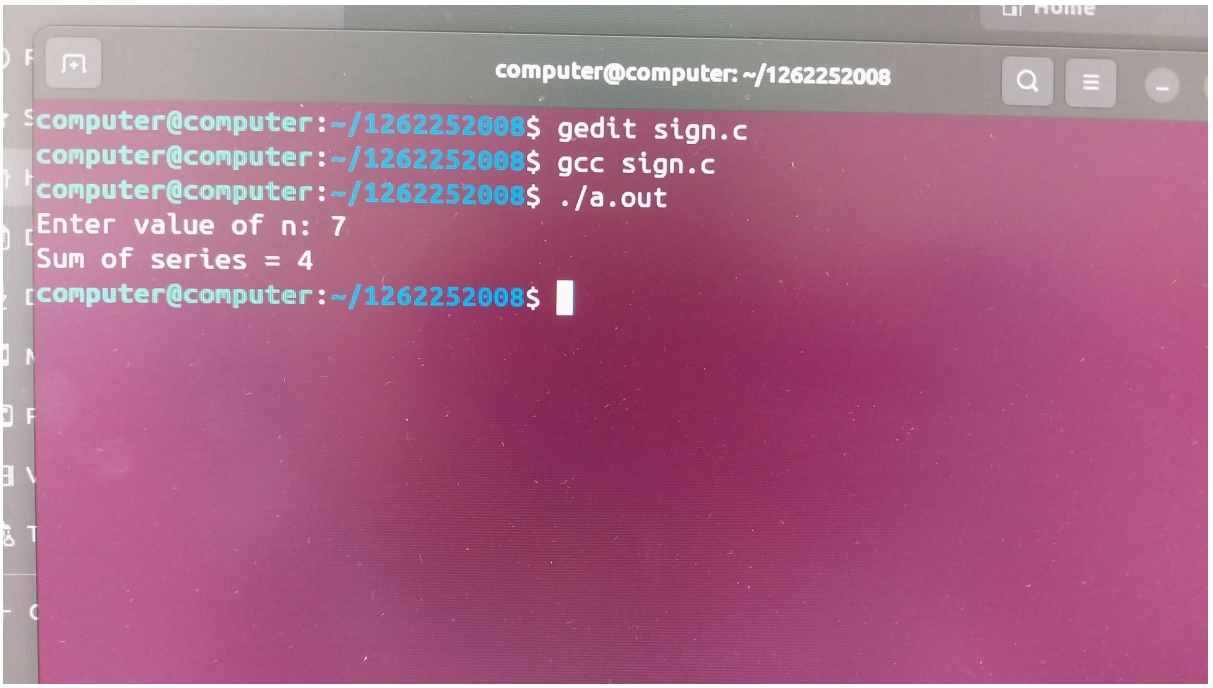


Assignment 17:

Sign series

```
#include <stdio.h>

int main() {
    int n, i;
    int sum = 0;

    printf("Enter value of n: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++) {
        if(i % 2 == 0)
            sum -= i;
        else
            sum += i;
    }

    printf("Sum of series = %d\n", sum);
```

```
    return 0;
}
```

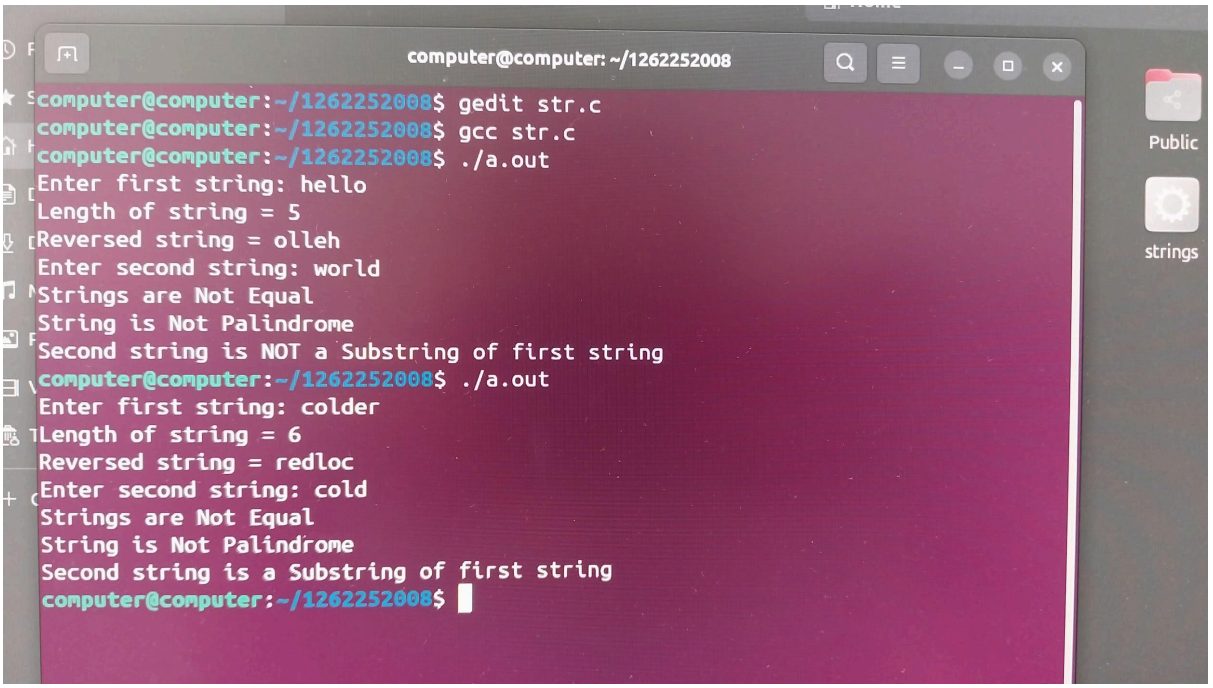


Assignment 18:

String operations

```
#include <stdio.h>
#include <string.h>
int main() {
char str1[100], str2[100], rev[100];
int i, len = 0, flag = 0;
printf("Enter first string: ");
scanf("%s", str1);

// i. Length of string
while(str1[len] != '\0')
len++;
printf("Length of string = %d\n", len);

// ii. String reversal
for(i = 0; i < len; i++) {
rev[i] = str1[len - i - 1];
}
rev[len] = '\0';
```

```
printf("Reversed string = %s\n", rev);

// iii. Equality check
printf("Enter second string: ");

scanf("%s", str2);
if(strcmp(str1, str2) == 0)
printf("Strings are Equal\n");
else
printf("Strings are Not Equal\n");

// iv. Palindrome check
if(strcmp(str1, rev) == 0)
printf("String is Palindrome\n");
else
printf("String is Not Palindrome\n");

// v. Substring check
if(strstr(str1, str2) != NULL)
printf("Second string is a Substring of first string\n");
else
printf("Second string is NOT a Substring of first string\n");
return 0;
}
```



Assignment 19:

Employee structure

```
#include <stdio.h>
#include <string.h>
struct employee {
 char name[50];
 char designation[50];
 char gender;
 char doj[20]; // Date of Joining
 float salary;
};
// Function to count male and female employees
void countGender(struct employee e[], int n) {
 int male = 0, female = 0, i;
 for(i = 0; i < n; i++) {
 if(e[i].gender == 'M' || e[i].gender == 'm')
 male++;
 else if(e[i].gender == 'F' || e[i].gender == 'f')
 female++;
 }
 printf("\nMale Employees: %d", male);
 printf("\nFemale Employees: %d\n", female);
}
// Function to display employees with salary > 10000
void highSalary(struct employee e[], int n) {
 int i;
 printf("\nEmployees with Salary > 10000:\n");

for(i = 0; i < n; i++) {
 if(e[i].salary > 10000) {
 printf("%s\n", e[i].name);
 }
 }
}
// Function to display employees with designation "Asst Manager"
void asstManager(struct employee e[], int n) {
 int i;
 printf("\nEmployees with designation 'Asst Manager':\n");
 for(i = 0; i < n; i++) {
 if(strcmp(e[i].designation, "Asst Manager") == 0) {
 printf("%s\n", e[i].name);
 }
 }
}
int main() {
 struct employee e[100];
```

```
 int n, i;
 printf("Enter number of employees: ");
 scanf("%d", &n);
 // Input details
 for(i = 0; i < n; i++) {
 printf("\nEnter details of employee %d\n", i + 1);
 printf("Name: ");

scanf("%s", e[i].name);
 printf("Designation: ");
 scanf("%s", e[i].designation);
 printf("Gender (M/F): ");
 scanf(" %c", &e[i].gender);
 printf("Date of Joining: ");
 scanf("%s", e[i].doj);
 printf("Salary: ");
 scanf("%f", &e[i].salary);
 }
 // a) Total employees
 printf("\nTotal number of employees: %d\n", n);
 // b) Count male and female
 countGender(e, n);
 // c) Salary > 10000
 highSalary(e, n);
 // d) Asst Manager
 asstManager(e, n);
 return 0;
}
```

Assignment 22:

String functions

```
#include <stdio.h>
#include <string.h>
int main() {
char str1[100], str2[100], temp[100];
int choice;
printf("Enter first string: ");
scanf("%s", str1);
printf("Enter second string: ");
scanf("%s", str2);
do {
printf("\n--- String Operations Menu ---\n");
printf("1. Length of string\n");
printf("2. Copy string\n");
```

```
printf("3. Concatenate strings\n");
printf("4. Compare strings\n");
printf("5. Reverse string\n");
printf("6. Check Palindrome\n");
printf("7. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch(choice) {
case 1:
printf("Length of str1 = %lu\n", strlen(str1));
break;
case 2:
strcpy(temp, str1);
printf("Copied string = %s\n", temp);
break;
case 3:
strcpy(temp, str1);
strcat(temp, str2);
printf("Concatenated string = %s\n", temp);
break;
case 4:
if(strcmp(str1, str2) == 0)
printf("Strings are Equal\n");
else
printf("Strings are Not Equal\n");
break;
case 5:
strcpy(temp, str1);
strrev(temp); // Note: may not work in all compilers
printf("Reversed string = %s\n", temp);
break;
case 6:
strcpy(temp, str1);
strrev(temp);
if(strcmp(str1, temp) == 0)
printf("Palindrome string\n");
else
printf("Not a palindrome\n");
break;
case 7:
printf("Exiting...");
break;
default:
printf("Invalid choice\n");
}
} while(choice != 7);
return 0;
}
```